

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Pseudocode Writing Task (Algorithm Design)

COURSE NAME: COMPUTER VISION WITH OPENCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: *Understand video processing - motion computation - 3D vision - geometry.* (BTL 6)

Assessment Tool Weightage: 20%

Pseudocode Writing Tasks: Design a clear and structured pseudocode for the given problem by organizing the solution into logical stages of a computer vision pipeline. Ensure proper use of control flow, modular steps, and justification of key operations to satisfy the given constraints.

1. Real-Time Face Detection and Recognition Pipeline

A system must process a live video stream to:

- A. Capture frames from a camera
- B. Detect faces in each frame
- C. Recognize faces using a pre-trained dataset
- D. Display bounding boxes and labels in real time

Constraints:

- Must operate continuously
- Should minimize processing delay
- Should handle multiple faces

Write detailed pseudocode for the complete pipeline using OpenCV concepts, including image acquisition, preprocessing, detection, recognition, and display. Ensure modular design and efficient looping structure.

2. OCR Pipeline for Document Processing

A document processing system must:

- A. Read input images containing printed text

- B. Enhance image quality (noise removal, contrast improvement)
- C. Segment text regions
- D. Extract and display recognized text

Constraints:

- Input images may be noisy and low contrast
- Batch processing required

Write pseudocode for an end-to-end OCR pipeline using OpenCV functions. Clearly structure preprocessing, segmentation, and text extraction stages.

3. Motion Detection and Tracking System

A surveillance system must:

- A. Read video input
- B. Detect motion between consecutive frames
- C. Track moving objects
- D. Highlight detected motion regions

Constraints:

- Should handle dynamic background
- Must operate in near real-time

Design pseudocode for the system, including frame differencing, filtering, tracking logic, and visualization. Ensure proper control flow and handling of continuous video streams.

4. Vehicle Counting System using Video

A traffic monitoring system must:

- A. Read video frames
- B. Detect vehicles entering a region of interest (ROI)
- C. Track movement across frames
- D. Count vehicles and display count

Constraints:

- Avoid double counting
- Handle multiple vehicles simultaneously

Write pseudocode for the complete system, including detection, tracking, counting logic, and visualization using drawing functions.

5. Facial Expression Recognition System

A mobile-based system must:

- A. Capture images from a camera

- B. Detect faces
- C. Extract facial features
- D. Classify expressions (happy, sad, neutral)
- E. Display results

Constraints:

- Limited computational resources
- Real-time response required

Write pseudocode for the pipeline, ensuring efficient processing and modular design for detection, feature extraction, and classification.

6. Real-Time Object Detection Pipeline

A system must:

- A. Capture video frames
- B. Detect objects using a trained model
- C. Draw bounding boxes and labels
- D. Display results in real time

Constraints:

- Must process frames efficiently
- Handle multiple objects

Write detailed pseudocode with modular steps for detection, visualization, and efficient looping.

7. Image Difference Detection System

A system must:

- A. Read two images
- B. Compare them
- C. Highlight differences

Constraints:

- Handle noise
- Ensure accurate difference detection

Write pseudocode including preprocessing, comparison logic, and visualization.

8. Video Frame Extraction System

A system must:

- A. Read video input
- B. Extract frames at fixed intervals
- C. Save selected frames

Constraints:

- Efficient storage
- Avoid redundant frames

Write structured pseudocode for frame selection and saving logic.

9. Background Subtraction Pipeline

A surveillance system must:

- A. Read video stream
- B. Separate foreground from background
- C. Highlight moving objects

Constraints:

- Handle dynamic background
- Reduce noise

Write pseudocode including background modeling, subtraction, filtering, and display.

10. Edge Detection Pipeline for Real-Time Video

A system must:

- A. Capture video frames
- B. Apply edge detection
- C. Display results

Constraints:

- Real-time processing
- Maintain clarity of edges

Write pseudocode with preprocessing, edge detection, and visualization modules.

11. Multi-Image Batch Processing System

A system must:

- A. Read multiple images from a folder
- B. Apply preprocessing and feature detection
- C. Store results

Constraints:

- Batch processing
- Efficient memory usage

Write pseudocode with modular structure and looping logic.

12. Face Detection with ROI Extraction

A system must:

- A. Detect faces in an image
- B. Extract face regions
- C. Save cropped faces

Constraints:

- Handle multiple faces
- Avoid redundant processing

Write pseudocode including detection, ROI extraction, and saving logic.

13. Video Stabilization Pipeline

A system must:

- A. Read video frames
- B. Detect motion between frames
- C. Stabilize output

Constraints:

- Maintain real-time processing
- Reduce jitter

Write pseudocode including motion estimation, transformation, and display.

14. Text Detection in Natural Scene Images

A system must:

- A. Read images
- B. Detect text regions
- C. Highlight detected areas

Constraints:

- Handle varying backgrounds
- Improve detection accuracy

Write pseudocode including preprocessing, region detection, and visualization.

15. Feature Matching Pipeline

A system must:

- A. Read two images
- B. Extract features
- C. Match features
- D. Display matches

Constraints:

- Robust to scale and rotation
- Efficient matching

Write pseudocode with modular steps for detection, matching, and visualization.

Code of Conduct:

I Bhavya PS (192524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach

Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1. Face Detection with ROI Extraction

Objective: Detect multiple faces, extract their Region of Interest (ROI), and save each cropped face without processing the same face repeatedly.

Pseudocode

START

Read input image

Convert image to suitable format

Load face detection model

Detect all faces in the image

Create an empty set for processed face regions

FOR each detected face:

Get face coordinates

x, y, width, height

Check whether this face was already processed

IF face is not already processed:

Extract face ROI using coordinates

Resize ROI if required

Generate a unique filename

Save cropped face

Add face coordinates to processed set

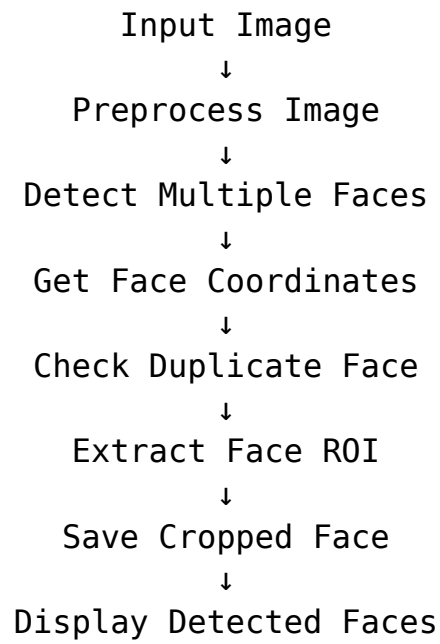
END IF

END FOR

Display original image with face rectangles

STOP

Flow



Result: Each detected face is extracted and saved separately while avoiding unnecessary repeated processing.

2. Video Stabilization Pipeline

Objective: Read video frames, estimate motion between consecutive frames, calculate the required transformation, and produce a stabilized video while reducing jitter.

Pseudocode

START

Open input video

Read video properties

Set required frame rate

Read first frame

Convert first frame to grayscale

Initialize previous frame

WHILE video has frames:

Read current frame

Convert current frame to grayscale

Detect feature points in previous frame

Track corresponding points in current frame

Estimate motion between frames

Calculate transformation matrix

Smooth the estimated transformation
to reduce sudden camera movement

Apply transformation to current frame

Display stabilized frame

Save stabilized frame if required

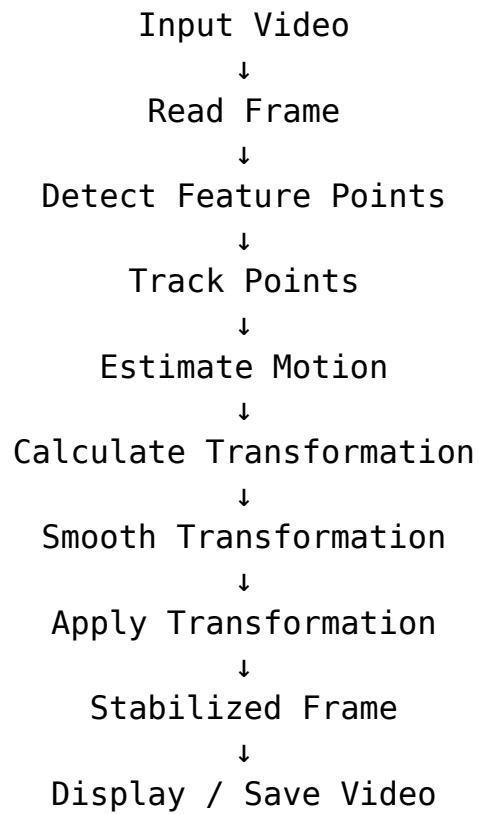
Set current frame as previous frame

END WHILE

Release video
Close display window

STOP

Flow



Result: The sudden unwanted camera movements are reduced, producing a smoother and more stable video.

3. Text Detection in Natural Scene Images

Objective: Detect text regions in images with different backgrounds, improve detection accuracy through preprocessing, and highlight the detected text.

Pseudocode

START

Read input image

Resize image if required

Convert image to grayscale

Reduce image noise

Improve contrast

Apply suitable thresholding or edge detection

Detect possible text regions

FOR each detected region:

Check region size and shape

Remove regions that are unlikely to contain text

Apply additional filtering if required

END FOR

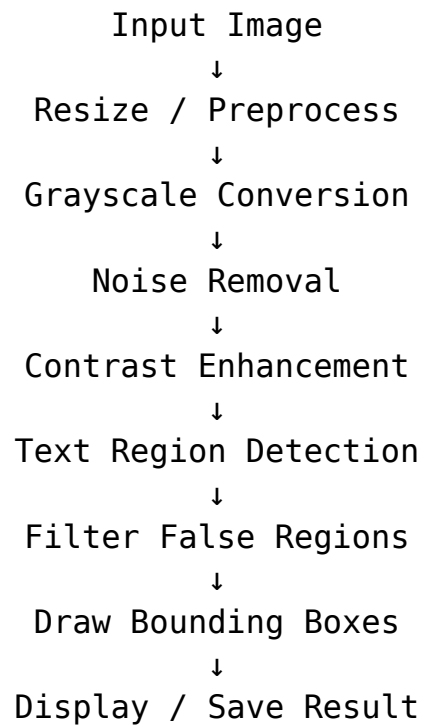
Draw bounding boxes around detected text regions

Display highlighted image

Save result if required

STOP

Flow



Result: Text areas are detected and highlighted even when the image contains different backgrounds, while preprocessing and filtering help reduce false detections.